



Short Communication

Fat loss and muscle gain: The possible role of striatal dopaminergic tone in determining the efficacy of physical exercise

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ABSTRACT

The effectiveness of exercise for obesity is contentious due to individual response variability. Owing to the roles of dopamine in motor functions, metabolism, and appetite, this study aimed to identify striatal dopamine as a predictor of variability in exercise response, specifically in terms of fat loss and muscle gain. Healthy non-exercising males completed an 8-week program, exercising 1 h, 4 days a week. Striatal dopaminergic tone was assessed by measuring dopamine transporter availability using technetium-99 m labelled tropine derivative, [^{99m}Tc]TRODAT-1 (TRODAT), single-photon emission computed tomography, and body composition (fat and muscles mass) was analysed using bioelectrical impedance. Lower baseline dopamine levels were associated with greater fat mass loss ($r = 0.58$, $p = 0.006$), percentage fat mass loss ($r = 0.53$, $p = 0.013$), and increase in muscle mass ($\beta = -0.53$, $p = 0.035$, after taking age and smoking status as covariates). These findings enhance our understanding of obesity neurobiology and exercise response variability, necessitating further research for targeted interventions based on dopaminergic profiles.

1. Introduction

The intrinsic effect of exercise on obesity is contentious [1–3]. While some individuals experience significant reductions in body fat and other outcomes, others show minimal changes [1–3]. Nevertheless, the specific molecular mechanisms contributing to the variability of exercise programs in terms of changes in body fat or muscle mass remain unknown.

Dysfunctions in the dopaminergic system can affect appetite [4] and various metabolic processes, including glucose and lipid metabolism, disrupting metabolic rate and energy consumption [5]. Dopamine may modulate metabolic biomarkers such as insulin, insulin sensitivity, HbA1c, and lipids [6,7]. Antidopaminergic medications for various psychological conditions accompany metabolic complications such as diabetes, obesity, appetite, hypertension, hyperglycemia, and dyslipidemia due to abnormal glucose and lipid metabolism [8,9]. These

outcomes linked to dopaminergic modulations may impact the efficacy of physical exercise on fat loss and other outcomes [3].

Studies have reported that dopaminergic system exhibits a compensatory response to exercise programs [10,11]. In diseases such as Parkinson's, where the dopaminergic system is compromised, exercise enhances basal ganglia dopamine levels, dendritic spine growth, and neuronal density more in diseased mice than in healthy counterparts [10–12]. Additionally, measures such as motor behavior, muscle fitness, and coordination exhibited more significant enhancements in the Parkinson's mouse group than in the control group [12]. Similarly, human research indicates that physical exercise induces more positive behavioral changes (i.e., executive function and attention) related to attenuated dopaminergic tone in attention deficit/hyperactivity disorder and in individuals with lower cognitive performance compared to healthy counterparts [13–16]. However, the associations between dopamine and physical exercise outcomes (e.g., muscle gain, body fat loss, etc.)

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have not been confirmed in humans.

Considering the roles of dopamine in metabolism and energy consumption, as well as the influence of physical exercise on dopaminergic tone and compensatory response in individuals with low dopaminergic tone, we hypothesize that baseline dopaminergic tone before an extended exercise program may determine the variability in fat loss and muscle gain responses. In essence, individuals with diverse baseline dopaminergic tones may exhibit varying degrees of response to the exercise program in terms of fat loss and muscles gain.

2. Methods

2.1. Ethics statement

This study received ethical approval (IRB number: B-BR-110-040) from the Ethical Committee for Human Research at National Cheng Kung University Hospital. Before any procedures, written informed consent was obtained from all participants.

2.2. Participants

Healthy male participants, confirmed as mentally healthy by a senior psychiatrist using the Mini International Neuropsychiatric Interview (MINI), and medication-free for the past 6 months, were recruited. They demonstrated physical fitness without metabolic, pulmonary, or cardiovascular complications, maintaining stable vital signs. All were right-handed, non-exercising, engaging in less than 3 h of moderate-intensity exercise weekly.

2.3. Study design and exercise intervention

All participants underwent single-photon emission computed tomography (SPECT) imaging at baseline to assess their striatal DAT availability. Participants' body composition (body fat mass, percentage body fat mass, and skeletal muscle mass) were assessed at baseline and after an 8-week exercise program to evaluate the effects of exercise on changes in body composition.

Following American College of Sports Medicine guidelines, participants engaged in supervised exercise sessions lasting approximately 1 h, four times a week for 8 weeks. Each session included aerobic treadmill activity and circuit-type resistance training. The 1-h regimen comprised a 5–10 min full-body warm-up, 20 min of aerobic activity, 20 min of resistance training, and the remaining time dedicated to whole-body stretching and cooling.

Individual exercise intensity was determined using heart rate (HR) monitors ($57\text{--}63\% \pm 6$ beats of maximum heart rate [HRmax]) and perceived exertion (RPE) on a Borg scale of 9–11 (indicating light intensity) for the initial three weeks. Subsequently, the intensity was adjusted to a moderate level ($64\text{--}76\%$ of HRmax and RPE of 12–13) for the remaining weeks. The resistance training encompassed nine exercises targeting major muscle groups, including chest, biceps, triceps, shoulders, hamstrings, quadriceps, and calves.

2.4. Technetium-99m labelled tropane derivative, [^{99m}Tc]TRODAT-1 (TRODAT) single-photon emission computed tomography (SPECT)

The procedures and methods for SPECT and radioligand binding ratio analysis are detailed in our previous study [17]. In summary, participants were injected intravenously with 740 MBq (20 mCi) of TRODAT. Imaging began about 240 min post-injection, capturing SPECT images with a 360-degree rotation in 120 steps (50 s each) and utilizing a $128 \times 128 \times 16$ matrix.

2.5. Body composition analysis

Body composition was evaluated with an InBody multi-frequency

bioelectrical impedance device (InBody 520, Cerritos, CA, USA). During analysis, participants removed metal devices and excess clothing, retaining only exercise attire. Standing barefoot on the device with foot-soles touching electrodes, participants gripped handles with extended elbows, maintaining a static posture throughout the analysis.

2.6. Statistical analyses

Data analysis used SPSS software version 26. We employed the Shapiro-Wilk test and analysed skewness and kurtosis statistics to check the normality of the data. Except for skeletal muscle mass change, the remaining data were normally distributed. Pearson's correlation was utilized to find the correlations between DAT availability from SPECT and body composition measures from InBody, such as changes in body fat mass or percentage body fat. In contrast, for skeletal muscle mass, which had a non-normal distribution, Spearman's correlation was applied. Significance threshold was set at $p < 0.05$. Age and cigarettes smoking has been well known to affect the dopamine transports availability. [18,19] We included age and smoking status as covariates in a regression model. Furthermore, we conducted a series of sensitivity tests using DAT availability as the predictor and the confounders (cognition score (full IQ), and biomarkers of lipid and glucose metabolism (blood triglycerides and HbA1C levels)) as covariates. Due to the small sample size limiting the feasibility of a model to control for all potential confounders at once, we conducted a series of regression analyses. In these analyses, the confounders were included as covariates one by one.

3. Results

Thirty healthy participants, mean age 30.43 years (SD = 8.41) and mean education level 15.59 years (SD = 1.71), were initially enrolled. Of these, 21 (mean age = 30.14 years, SD = 9.48) completed the 8-week exercise program.

3.1. Dopamine transporter (DAT) availability at baseline associations with the changes in body fat mass, percentage body fat, and muscles mass

The change (Δ) was defined as visit 2 – visit 1 (i.e., Δ body fat mass = body fat mass at visit 2 - body fat mass at visit 1). Pearson's correlations showed that the basal ganglia's (bilateral striatum) DAT availability at baseline was significantly correlated with the change in body fat mass ($r = 0.58, p = 0.006$, Fig. 1A) and the change in percentage body fat ($r = 0.53, p = 0.013$, Fig. 1B) after 8 weeks of exercise program. This suggests that individuals with low DAT availability exhibit a greater decrease in body fat mass or percentage body fat. Taking age and smoking status as covariates in regression model, associations remained significant ($\beta = 0.47, t = 2.25, p = 0.038$ and $\beta = 0.51, t = 2.22, p = 0.04$, respectively). Furthermore, the associations between DAT availability and change in body fat mass or percentage body fat remained significant after controlling for a series of confounders one by one: blood triglyceride levels ($\beta = 0.59, t = 2.93, p = 0.01$ and $\beta = 0.55, t = 2.67, p = 0.02$), HbA1C ($\beta = 0.58, t = 2.95, p = 0.01$ and $\beta = 0.53, t = 2.58, p = 0.02$) and cognitive indicator (full-scale IQ, $\beta = 0.62, t = 3.29, p = 0.004$ and $\beta = 0.57, t = 2.87, p = 0.01$, respectively).

Contrastingly, Spearman's correlation showed an inverse association between DAT availability and muscle mass change, though initially nonsignificant ($\rho = -0.08, p = 0.73$, Fig. 1C), became significant after adjusting for age and smoking status as covariates in a regression model ($\beta = -0.53, t = -2.29, p = 0.035$). This suggests that individuals with low DAT availability manifest heightened muscle gain. However, after controlling for the effects of metabolic and cognitive markers such as blood triglyceride levels, HbA1C, and full-scale IQ, either in a fully adjusted model (taking all covariates at once) or one by one in a series of regressions, statistical significance was not attained (all $ps > 0.05$).

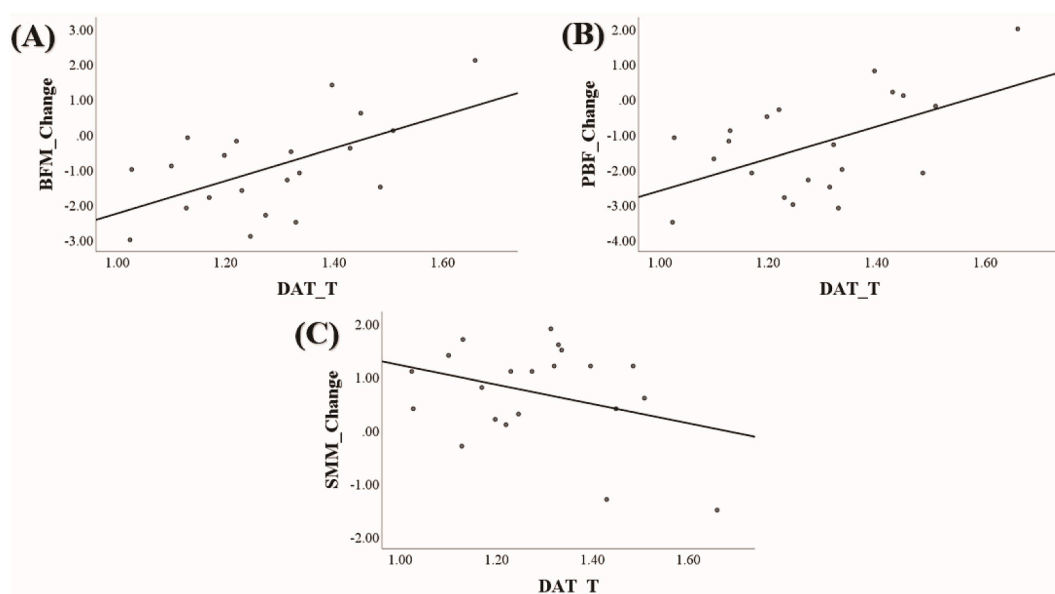


Fig. 1. Associations between baseline basal ganglia (bilateral striatum) DAT availability and changes in body fat mass (BFM_Change) (A), percentage body fat (PBF_Change) (B), and skeletal muscle mass (SMM_Change) (C) after 8 weeks of physical exercise

Abbreviations: BFM, body fat mass; PBF, percentage body fat; SMM, skeletal muscles mass; DAT, dopamine transporter; T, total (right + left striatum).

4. Discussion

This pioneering study examines how baseline striatal dopaminergic tone predicts post-exercise fat loss and muscle gain in humans. Our results suggest that lower baseline dopaminergic tone in healthy non-exercisers may contribute to increased fat loss and greater muscle gain post-physical exercise.

Studies have indicated that populations with diminished dopaminergic neuronal density or dopamine levels may experience a more significant post-exercise dopaminergic increase [10–12]. Similarly cognitive and executive functioning, which are associated with dopamine, studies have shown that individuals with lower cognitive functioning are more susceptible to benefit from exercise [13,15]. Research on physical activity, encompassing motor behavior, muscle fitness, and coordination, have reported greater improvements in dopamine-deficient populations, such as Parkinson's mice, compared to control groups [12]. Our study's findings align with prior research, supporting superior outcomes in populations with dopamine deficiencies and related disorders [10], – [12,14,15] possibly due to dopamine compensation.

Dopamine plays crucial roles in food addiction [4] and lipid and glucose metabolism [7]. Lower dopamine levels are linked to increased food intake [4]. Previous studies have highlighted dopamine's involvement in modulating metabolism [5,6]. It is plausible that non-exercising individuals with low dopaminergic tone may have a higher baseline appetite level and a lower metabolism, contributing to elevated body fat [4,5]. However, when these individuals begin to exercise regularly, it may lead to an increase in dopamine levels and a decrease in appetite [3], potentially resulting in a greater decrease in body fat over time. Unfortunately, we lack the essential data on appetite levels before and after the exercise program to empirically test this hypothesis.

Physical inactivity is associated with low dopamine, and an increase in physical activity correspondingly increases neurotransmitter activity, particularly dopamine. The increased gains observed after 8 weeks of physical exercise in healthy individuals who were previously non-exercisers and have low basal dopamine levels may also be explained in terms of neuromodulation [20,21]. It is a notion that beginners with no prior physical activity tend to respond more strongly to exercise in terms of energy expenditure, muscle gain, and strength acquisition [21].

The initial heightened response of energy expenditure, muscle and strength gains mainly involves neuronal adaptation [20,21]. Our findings may offer mechanistic insights into the heightened gains in beginners with low physical activity levels [21]. Our results suggest that individuals with low physical activity which is associated with low dopamine levels [22] may be more susceptible to significant changes in dopamine levels and subsequent alterations in body composition.

In summary, our findings suggest that variations in response to physical exercise, concerning fat loss and muscle gain, may stem from differences in dopamine levels. These findings enhance our understanding of obesity's neurobiology, underscoring dopamine's pivotal role in controlling metabolism and body composition.

There are several limitations of the current study. First, the sample size was small. Including covariates in regression models one by one maintained statistical power, particularly with changes in body fat mass or percentage body fat. However, the small sample size limited statistical power when multiple covariates were included simultaneously in fully adjusted models. Secondly, bioelectrical impedance technology for measuring body composition is susceptible to hydration levels and body temperature, that were not tightly controlled. Thirdly, while we have measured the overall change in body fat mass or percentage body fat, we lack specific data on fat distribution in particular body parts or in the muscles targeted by exercise.

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Data availability statement

The data used to support the findings of this study are available from the corresponding author upon request.

Declaration of competing interest

The funding institutions had no further role in the study design, the collection, analysis, and interpretation of data, the writing of this paper, or the decision to submit it for publication. The authors report no financial relationships with commercial interests.

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